Robust Parameter Design

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Introduction

Robust parameter design (Taguchi, 1950s-60s) seeks combinations of controllable design factors that minimise response variability across noise factors (environmental, batch-to-batch, use-conditions). The goal is a product or process that performs consistently regardless of noise.

Prerequisites

Factorial designs; variance decomposition.

Theory

Inner array of control factors crossed with **outer array** of noise factors. At each control setting, compute a summary – mean, standard deviation, or signal-to-noise ratio. Choose the control setting that optimises the summary.

Modern approach: fit a response model with control, noise, and control-x-noise interactions; find control levels that minimise variance while meeting the target mean.

Assumptions

Noise factors represent realistic variation at use time; interactions with control factors are the primary lever for variance reduction.

R Implementation

```
library(FrF2)

set.seed(2026)
# Inner array: 2^3 controllable factors
ctrl <- FrF2(nruns = 8, nfactors = 3, randomize = FALSE)
ctrl$A <- as.numeric(as.character(ctrl$A))
ctrl$B <- as.numeric(as.character(ctrl$B))
ctrl$C <- as.numeric(as.character(ctrl$C))

# Outer array: 2^2 noise factors
noise <- expand.grid(N1 = c(-1, 1), N2 = c(-1, 1))

# Simulate: noise x control interaction
results <- do.call(rbind, lapply(1:nrow(ctrl), function(i) {
  vals <- sapply(1:nrow(noise), function(j) {
    10 + 2 * ctrl$A[i] + 1.5 * ctrl$B[i] +
      0.5 * noise$N1[j] + 0.8 * ctrl$A[i] * noise$N1[j] +
      rnorm(1, 0, 0.3)
  })
  data.frame(run = i, mean = mean(vals), sd = sd(vals),
```

```r
              snr = -10 * log10(var(vals)))
}))
results

# Choose control level minimising variance
results$A <- ctrl$A
cbind(ctrl, results)[order(results$sd), ]
```

Output & Results

Per-control-run mean, SD, and signal-to-noise ratio; the control combination with the smallest SD is the robust-optimal.

Interpretation

"Robust parameter design identified setting (A = -1, B = +1, C = +1) as minimising response SD across noise factors. Control factor A interacted with noise factor N1, explaining the variance reduction at A = -1."

Practical Tips

- The inner-outer array approach can be computationally expensive; use a combined model with control-noise interactions when possible.
- Report both mean and variance; robust designs often accept a small mean-response penalty for large variance reduction.
- Consider dual-response surface methods for joint mean-variance optimisation.
- Pay attention to control-noise interactions: that is where robustness lives.
- Combining Taguchi's philosophy with modern RSM is more flexible than pure Taguchi methods.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Observational designs and STROBE
- Randomised controlled trials
- Adaptive, non-inferiority, equivalence trials
- Bench and translational design
- MCAR, MAR, MNAR
- Multiple imputation with mice
- Linear mixed models with lme4
- GLMMs and GEE
- DAGs with dagitty and ggdag
- Propensity scores and IPTW
- G-methods, IV, DiD, RDD; heterogeneity of treatment effect
- SIR and SEIR models with deSolve
- Pre-registration and statistical analysis plans